

Food Ordering System – Database Description

A **Food Ordering System** is a database application that allows customers to browse restaurants, view food menus, place orders, make payments, and track deliveries. The system stores information about customers, restaurants, menu items, orders, payments, and delivery personnel. It ensures that all food ordering activities are managed efficiently while maintaining accurate and consistent data.

Identity Entities, Attributes, Primary Keys, and Relationships

1. Customer

Primary Key (PK): CustomerID

Attributes:

- CustomerID
- CustomerName
- Email
- PhoneNumber
- Address
- Password
- RegistrationDate

Relationship:

- One customer can place many orders.
- **Relationship:** Customer (1) → (M) Order

2. Restaurant

Primary Key (PK): RestaurantID

Attributes:

- RestaurantID
- RestaurantName
- Address
- PhoneNumber
- CuisineType
- Rating

Relationship:

- One restaurant offers many menu items.
- **Relationship:** Restaurant (1) → (M) Menu

3. Menu

Primary Key (PK): MenuID

Foreign Key (FK): RestaurantID

Attributes:

- MenuID
- RestaurantID
- ItemName
- Description
- Category
- Price
- Availability

Relationship:

- One menu item can appear in many order items.
- **Relationship:** Menu (1) → (M) OrderItem

4. Order

Primary Key (PK): OrderID

Foreign Key (FK): CustomerID

Attributes:

- OrderID
- CustomerID
- OrderDate
- TotalAmount
- OrderStatus

Relationship:

- One customer places many orders.
- One order contains many order items.
- One order has one payment.
- One order has one delivery.

5. OrderItem

Primary Key (PK): OrderItemID

Foreign Keys (FK):

- OrderID
- MenuID

Attributes:

- OrderItemID
- OrderID
- MenuID
- Quantity
- Price

Relationship:

- Many order items belong to one order.
- Many order items refer to one menu item.

6. Payment

Primary Key (PK): PaymentID

Foreign Key (FK): OrderID

Attributes:

- PaymentID
- OrderID
- PaymentMethod
- PaymentDate
- Amount

- PaymentStatus
- TransactionID

Relationship:

- One payment belongs to one order.
- **Relationship:** Order (1) → (1) Payment

7. Delivery

Primary Key (PK): DeliveryID

Foreign Keys (FK):

- OrderID
- DeliveryBoyID

Attributes:

- DeliveryID
- OrderID
- DeliveryAddress
- DeliveryDate
- DeliveryStatus
- DeliveryBoyID

Relationship:

- One delivery belongs to one order.
- One delivery is assigned to one delivery person.

8. DeliveryBoy

Primary Key (PK): DeliveryBoyID

Attributes:

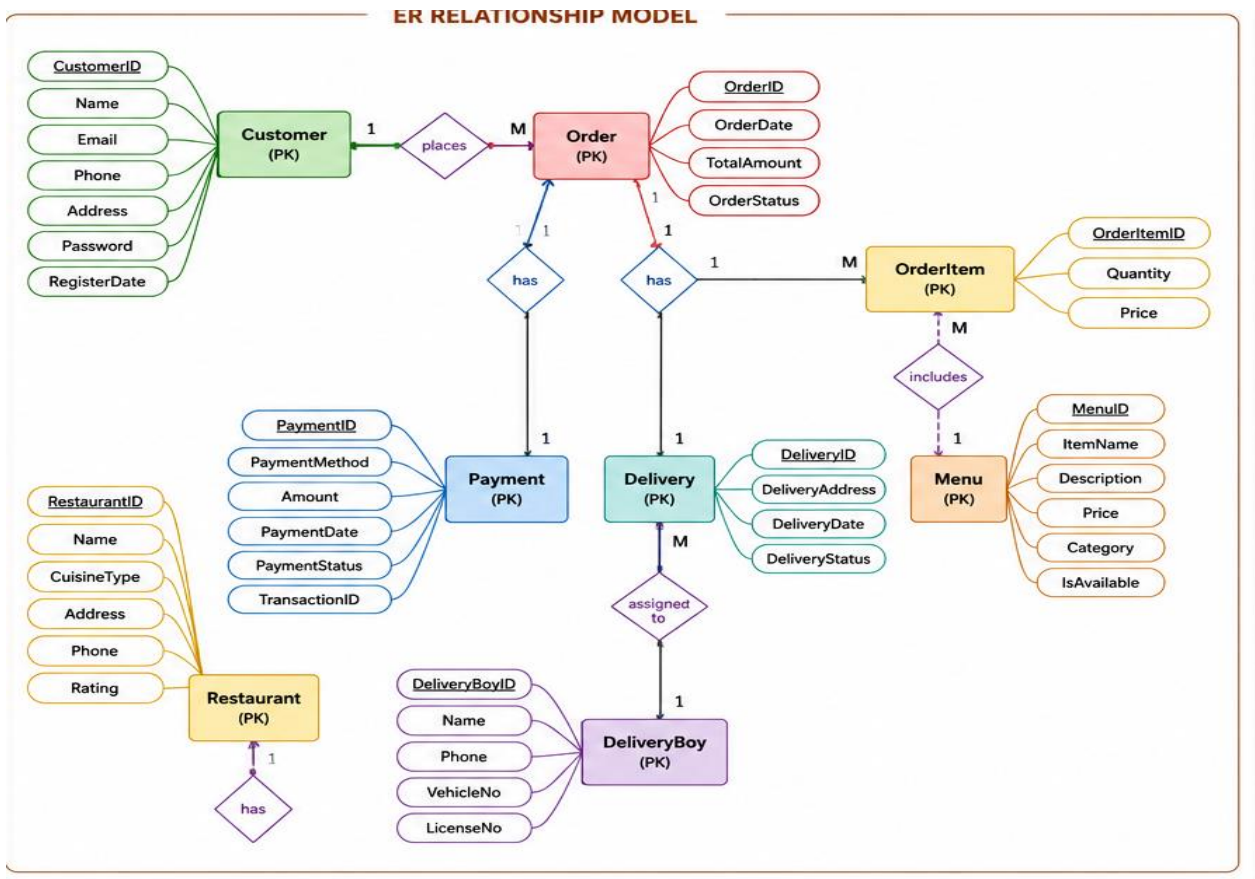
- DeliveryBoyID
- Name
- PhoneNumber
- VehicleNumber

- LicenseNumber

Relationship:

- One delivery person delivers many orders.
- **Relationship:** DeliveryBoy (1) → (M) Delivery

ER Relationship Model:



1. Customer places Order

- One customer can place many orders.
- **Cardinality:** 1 : M

2. Restaurant has Menu

- One restaurant can have many menu items.
- **Cardinality:** 1 : M
- 3. **Order contains OrderItem**
 - One order can contain many order items.
 - **Cardinality:** 1 : M
- 4. **Menu is included in OrderItem**
 - One menu item can appear in many order items.
 - **Cardinality:** 1 : M
- 5. **Order has Payment**
 - Each order has exactly one payment.
 - **Cardinality:** 1 : 1
- 6. **Order has Delivery**
 - Each order has one delivery record.
 - **Cardinality:** 1 : 1
- 7. **DeliveryBoy delivers Delivery**
 - One delivery person can deliver many orders.
 - **Cardinality:** 1 : M

Summary of Relationships

From Entity	Relationship	To Entity	Cardinality
Customer	Places	Order	1 : M
Restaurant	Has	Menu	1 : M
Order	Contains	OrderItem	1 : M

From Entity	Relationship	To Entity	Cardinality
Menu	Included In	OrderItem	1 : M
Order	Has	Payment	1 : 1
Order	Has	Delivery	1 : 1
DeliveryBoy	Delivers	Delivery	1 : M